VIT-AP UNIVERSITY

Amaravati, Andhra Pradesh, India

School of Computer Science and Engineering

Project Report

FITPULSE

Comprehensive Software Engineering Project Documentation

Course	Software Engineering
Assessment Component	7 - Project Report
Maximum Marks	10
Allocated Marks	5
Submission Date	14 April 2026
Project Type	Cloud-enabled Fitness Tracking Web Application
Latest Baseline Commit	8f7f44f
Live Production Link	https://fitpulse-swe.vercel.app/

GitHub Repository Link	https://github.com/abhinavsai2006/fitpulse

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1. Abstract

FitPulse is a modern fitness tracking web application designed to provide users with a unified platform for workouts, activity insights, goal planning, and profile management. The project evolved from a UI-focused implementation to a cloud-integrated system using Firebase Authentication and Firestore for secure account and data handling.

2. Problem Statement

Users often manage fitness data across disconnected tools, creating inconsistent tracking and low engagement. FitPulse addresses this by offering a single responsive application where authentication, progress monitoring, and account personalization are consolidated into one workflow.

3. Objectives and Scope

3.1 Objectives

- Deliver a clean and responsive fitness platform with modular frontend design.
- Implement robust authentication including verification and recovery workflows.
- Persist user profile and onboarding information securely in cloud storage.
- Achieve deployment readiness with reproducible environment configuration.

3.2 Scope

The delivered scope includes landing/auth flows, onboarding, dashboard, workouts, activity, nutrition, analytics, goals, profile, and settings modules with Firebase-backed account management and Firestore persistence for user-centric data.

4. Development Methodology

The execution followed an iterative implementation cycle: module development, integration, validation, and correction. High-impact issues were prioritized first, especially authentication stability, secure configuration, and production deployment consistency.

Phase	Activities	Outcome
Planning and UI baseline	Module identification, page scaffolding, visual consistency	Functional frontend shell completed
Cloud integration	Firebase Auth and Firestore service integration	Persistent and secure user workflows enabled
Stabilization	Auth edge-case handling, env migration, rule deployment	Production reliability improved
Validation and reporting	Build checks, test-case verification, report finalization	Submission-ready artifact set produced

5. Requirements Analysis

5.1 Functional Requirements

ID	Requirement	Status
FR-01	User registration and sign-in with email/password	Implemented
FR-02	Email verification enforcement for password accounts	Implemented
FR-03	Google sign-in with popup and redirect fallback	Implemented
FR-04	Password reset workflow	Implemented
FR-05	Onboarding and profile persistence in Firestore	Implemented
FR-06	Dashboard and feature-page access control	Implemented
FR-07	Workout lifecycle state transitions	Implemented

5.2 Non-functional Requirements

ID	Requirement	Approach
NFR-01	Responsive and mobile-friendly UI	Adaptive layout behavior and mobile full-screen refinements
NFR-02	Security of configuration values	Environment variable based Firebase setup and gitignore controls
NFR-03	Data access control	Owner-only Firestore security rules
NFR-04	Build and deployment reliability	Vite build checks and Vercel production redeploy verification

ID	Requirement	Approach
NFR-05	Maintainable architecture	Context-reducer pattern and modular page-component separation

6. System Design and Architecture

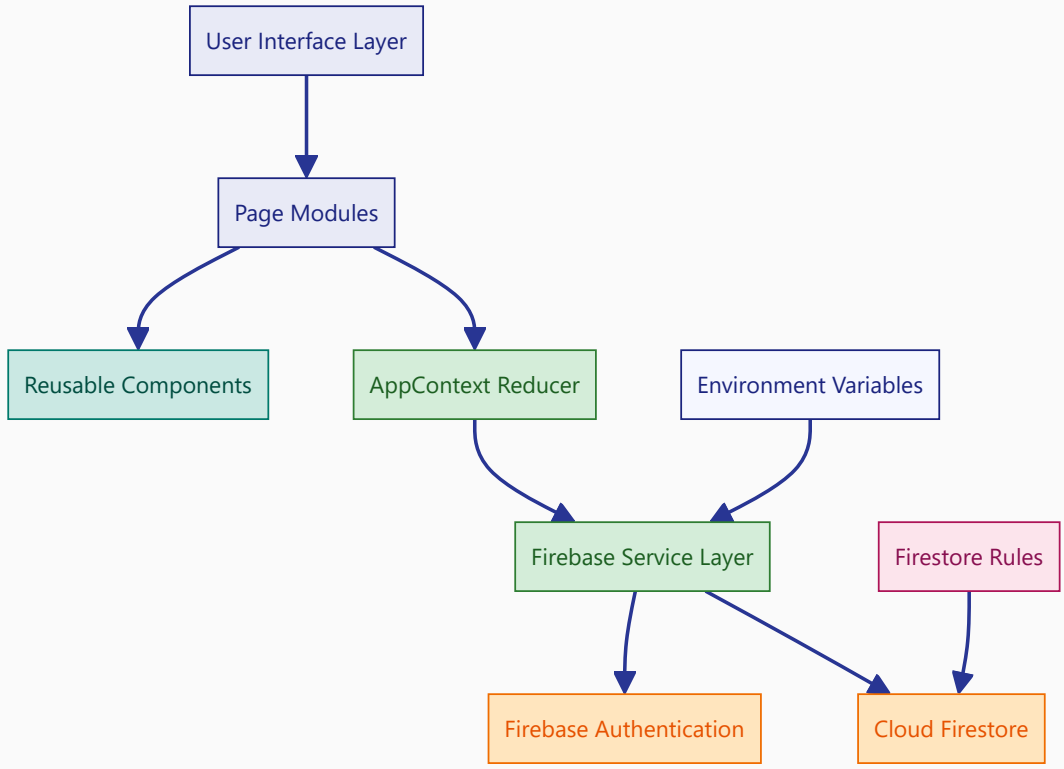

Figure 1: Updated FitPulse architecture with cloud services

6.1 Data and control flow

Authentication events are handled through AppContext, which synchronizes session state with Firebase and updates UI routing. Firestore acts as the source of truth for user profile and onboarding details. This design keeps business logic centralized and minimizes state drift between pages.

7. Implementation Summary

Area	Implemented Elements	Outcome
Authentication	Email/password login, verification checks, reset flow, Google auth fallback	Reliable account lifecycle support
State management	Reducer-driven auth, navigation, and workout transitions	Predictable behavior across modules
Persistence	Firestore profile creation, merge, and onboarding writes	Cloud-backed user continuity
Security hardening	Environment variable migration and owner-based rules	Safer production posture
Deployment	Vercel production redeploy and Firebase rules release	Live service stability

7.1 Module inventory

Category	Modules	Purpose
Entry layer	main.jsx, App.jsx	Bootstrapping and page-level control flow
State layer	AppContext.jsx	Global state, auth actions, and guarded transitions
Service layer	firebase.js	Firebase app initialization and profile utility operations
Feature pages	Landing, Auth, Onboarding, Dashboard, Workouts, Activity, Nutrition, Analytics,	User-facing workflows

Category	Modules	Purpose
	Goals, Profile, Settings	
Config assets	.env.example, firebase.json, firestore.rules	Secure configuration and cloud rule governance

8. Testing and Validation Summary

Validation Area	Method	Result
Authentication scenarios	Manual and flow-based test cases	Pass
Data persistence and profile sync	Firestore verification checks	Pass
Responsive behavior	Desktop and mobile viewport checks	Pass
Build stability	npm run build	Pass
Rules deployment	firebase deploy --only firestore:rules	Pass

9. Deployment, Security, and Operations

Operational Area	Action Taken	Status
Vercel environment setup	VITE_FIREBASE_* values configured for production and development	Completed
Configuration hygiene	Firebase values moved from source to environment variables	Completed
Firestore security	Owner-only read/write rules published to fitpulse-7173f	Completed
Production redeploy	Fresh deployment and alias verification on fitpulse-swe.vercel.app	Completed
Known limitation	Preview env in non-interactive CLI requires branch-scoped setup	Documented

10. Results and Discussion

The latest project baseline demonstrates meaningful progression from a presentation-focused prototype to an operational cloud-connected application. Authentication reliability improved through explicit verification handling and popup fallback logic, while data integrity improved through secured Firestore access and profile synchronization controls.

Outcome Parameter	Observation	Interpretation
Feature completeness	Core pages plus auth and profile persistence implemented	Primary project goals met

Outcome Parameter	Observation	Interpretation
Security posture	Environment-based config and owner-only rules active	Reduced risk of data exposure
Deployment readiness	Build and production deploy validated	Submission-ready runtime baseline
User experience stability	Auth flows and mobile layout behavior improved	Higher practical usability

11. Conclusion and Future Enhancements

FitPulse successfully fulfills the required software engineering outcomes across implementation, testing, documentation, and deployment verification. The project demonstrates practical engineering maturity through modular design, secure cloud integration, and structured validation.

11.1 Proposed future enhancements

- Role-based access controls and admin analytics.
- Notification and engagement engine with streak tracking.
- Advanced personalized recommendations from user activity history.
- Cross-platform expansion using shared design and service contracts.

12. Deployment and Repository Links

Resource	Link	Description
Live Production	https://fitpulse-swe.vercel.app/	Public validation endpoint
GitHub Repository	https://github.com/abhinavsai2006/fitpulse	Source, commits, and report assets

13. References

1. React Documentation - https://react.dev/
2. Vite Documentation - https://vite.dev/
3. Firebase Documentation - https://firebase.google.com/docs

4. Cloud Firestore Security Rules - https://firebase.google.com/docs/firestore/security/get-started
5. Vercel Documentation - https://vercel.com/docs

End of Document

Project Report - FitPulse
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